将 FlexVol 卷转换为 FlexGroup 卷

ONTAP 9

NetApp
January 27, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/flexgroup/convert-flexvol-concept.html on January 27, 2022. Always check docs.netapp.com for the latest.

目录

将 FlexVol 卷转换为 FlexGroup 卷 1
将 FlexVol 卷转换为 FlexGroup 卷概述 1
将 FlexVol 卷转换为 FlexGroup 卷 2
将 FlexVol 卷 SnapMirror 关系转换为 FlexGroup 卷 SnapMirror 关系 3

将 FlexVol 卷转换为 FlexGroup 卷

将 FlexVol 卷转换为 FlexGroup 卷概述

如果要将 FlexVol 卷扩展到超出其空间限制的范围，则可以将 FlexVol 卷转换为 FlexGroup 卷。从 ONTAP 9.7 开始，您可以将具有 SnapMirror 关系的独立 FlexVol 卷或 FlexVol 卷转换为 FlexGroup 卷。

将 FlexVol 卷转换为 FlexGroup 卷的注意事项

在决定将 FlexVol 卷转换为 FlexGroup 卷之前，您应了解支持的功能和操作。

转换期间不支持操作

在进行卷转换时，不允许执行以下操作：

- 卷移动
- 聚合自动执行
- 聚合重新定位
- 在高可用性配置中执行计划内接管和交还
- 在高可用性配置中手动和自动交还
- 集群升级和还原
- FlexClone 卷拆分
- 卷重新托管
- 卷修改和自动调整大小
- 卷重命名
- 将对象存储附加到聚合
- MetroCluster 配置中的协商切换
- SnapMirror 操作
- 从 Snapshot 副本还原
- 配额操作
- 存储效率操作

成功转换后，您可以对 FlexGroup 卷执行这些操作。

FlexGroup 卷不支持的配置

- 使卷脱机或受限
- SVM 根卷
- SnapLock 卷

- SAN
- SMB 1.0
- NVMe 命名空间
- 远程卷影复制服务（VSS）

将 FlexVol 卷转换为 FlexGroup 卷

从 ONTAP 9.7 开始，您可以将 FlexVol 卷原位转换为 FlexGroup 卷，而无需数据副本或额外的磁盘空间。

您需要什么？ **#8217**；将需要什么

- 过渡后的卷可以转换为 FlexGroup 卷。

  要将过渡后的卷转换为 FlexGroup 卷，需要使用 diag 模式，并且只能在 NetApp 支持部门的监督下执行。强烈建议您在继续进行转换过程之前联系 NetApp 支持部门。

- 要转换的 FlexVol 卷必须处于联机状态。
- FlexVol 卷上的操作和配置必须与转换过程兼容。

  如果 FlexVol 卷不兼容，并且卷转换已中止，则会生成一条错误消息。您可以采取更正操作并重试转换。

步骤

1. 验证 FlexVol 卷是否联机： `volume show vol_name -volume-style-Extended ， state`

```
cluster-1::> volume show my_volume -fields volume-style-extended,state
vserver volume    state  volume-style-extended
------- --------- ------ ---------------------
vs0     my_volume online flexvol
```

2. 验证是否可以无问题地转换 FlexVol 卷：

   a. 登录到高级权限模式： `set -privilege advanced`

   b. 验证转换过程： `volume conversion start -vserver vs1 -volume FlexVol -check -only true`

   在转换卷之前，必须更正所有错误。

   > 您不能将 FlexGroup 卷转换回 FlexVol 卷。如果要执行此操作，应联系支持部门。

3. 开始转换： `volume conversion start -vserver svm_name -volume vol_name`

```
cluster-1::*> volume conversion start -vserver vs0 -volume my_volume

Warning: Converting flexible volume "my_volume" in Vserver "vs0" to a
FlexGroup
         will cause the state of all Snapshot copies from the volume to
be set
         to "pre-conversion". Pre-conversion Snapshot copies cannot be
         restored.
Do you want to continue? {y|n}: y
[Job 57] Job succeeded: success
```

4. 验证转换是否成功： `volume show vol_name -fields -volume-style-Extended ， state`

```
cluster-1::*> volume show my_volume -fields volume-style-extended,state
vserver volume    state  volume-style-extended
------- --------- ------ ---------------------
vs0     my_volume online flexgroup
```

FlexVol 卷将转换为单成员 FlexGroup 卷。

您可以根据需要扩展 FlexGroup 卷。

将 FlexVol 卷 SnapMirror 关系转换为 FlexGroup 卷 SnapMirror 关系

要在 ONTAP 中将 FlexVol 卷 SnapMirror 关系转换为 FlexGroup 卷 SnapMirror 关系，必须先转换目标 FlexVol 卷，然后再转换源 FlexVol 卷。

您需要什么？ **#8217**；将需要什么

- 要转换的 FlexVol 卷必须处于联机状态。
- SnapMirror 关系中的源 FlexVol 卷不能是多个 SnapMirror 关系的源卷。

  FlexGroup 卷不支持扇出 SnapMirror 关系。

- FlexVol 卷上的操作和配置必须与转换过程兼容。

  如果 FlexVol 卷不兼容，并且卷转换已中止，则会生成一条错误消息。您可以采取更正操作并重试转换。

只有异步 SnapMirror 关系才支持 FlexGroup 转换。

步骤

1. 验证 SnapMirror 关系是否运行正常： `snapmirror show`

   只能转换 XDP 类型的镜像关系。

```
cluster2::> snapmirror show

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vs0:src_dpv DP   vs2:dst_dpv  Snapmirrored
                                      Idle           -         true    -
vs0:src_xdp XDP  vs2:dst_xdp  Snapmirrored
                                      Idle           -         true    -
```

2. 验证源卷是否与转换兼容：
   a. 登录到高级权限模式：`set -privilege advanced`
   b. 验证转换过程：`volume conversion start -vserver vs1 -volume src_vol -check -only true`

      在转换卷之前，必须更正所有错误。

3. 将目标 FlexVol 卷转换为 FlexGroup 卷。
   a. 暂停 FlexVol SnapMirror 关系：`snapmirror quiesce -destination-path dest_svm ：dest_volume`

      ```
      cluster2::> snapmirror quiesce -destination-path vs2:dst_xdp
      ```

   b. 开始转换：`volume conversion start -vserver dest_svm -volume dest_volume`

      ```
      cluster-1::> volume conversion start -vserver vs2 -volume dst_xdp

      Warning: After the volume is converted to a FlexGroup, it will not be
      possible
      to change it back to a flexible volume.
      Do you want to continue? {y|n}: y

      [Job 510] Job succeeded: SnapMirror destination volume "dst_xdp" has
      been successfully converted to a FlexGroup volume.
      You must now convert the relationship's source volume, "vs0:src_xdp",
      to a FlexGroup.
      Then, re-establish the SnapMirror relationship using the "snapmirror
      resync" command.
      ```

4. 将源 FlexVol 卷转换为 FlexGroup 卷： `volume conversion start -vserver src_svm_name -volume src_vol_name`

```
cluster-1::> volume conversion start -vserver vs0 -volume src_xdp

Warning: Converting flexible volume "src_xdp" in Vserver "vs0" to a
FlexGroup
         will cause the state of all Snapshot copies from the volume to
be set
         to "pre-conversion". Pre-conversion Snapshot copies cannot be
         restored.
Do you want to continue? {y|n}: y
[Job 57] Job succeeded: success
```

5. 重新同步此关系： `snapmirror resync -destination-path dest_svm_name ： dest_volume`

```
cluster2::> snapmirror resync -destination-path vs2:dst_xdp
```

您必须确保在扩展源 FlexGroup 卷以包含更多成分卷时，目标卷也会进行扩展。

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.